

Dual mapping — code walkthrough
CPU snippets (same role as N2S-CLI-Walkthrough.pdf)

cxr-evidence-grounding-lab/notes

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Dual mapping — code walkthrough

Same job as docs/N2S-CLI-Walkthrough.pdf: **Intuition** → **copy-paste Python** → **send-back**. CPU only. No GPU fill. Does not rewrite ground().

Lab root: ~/staging/cxr-evidence-grounding-lab

Python: ../cxrlabs/faiss_gpu1/bin/python

Not this PDF: mapping-notes-simple.pdf (GUI walk), CXR-Mapping-Curriculum.pdf (MAP-00...09), progress-notes.pdf (M1-Ph14).

C0 — Path setup

Intuition. Every later snippet assumes the lab is on sys.path and Dual-fill extracts already exist.

```
cd ~/staging/cxr-evidence-grounding-lab
ls artifacts/n2s-g2-analog-fill-extracts.json n2s_lab/ground.py n2s_lab/predicate.py
```

Send-back: C0 done plus the ls output.

C1 — Extract, ground, rule (no patch)

Intuition. The model is not in this loop. Frozen JSON → Extraction → ground() → evaluate_rule().

```
cd ~/staging/cxr-evidence-grounding-lab
../cxrlabs/faiss_gpu1/bin/python - <<'PY'
import json
from n2s_lab.n2s_g2_analog_fill import EXTRACT_PATH
from n2s_lab.ground import ground
from n2s_lab.predicate import evaluate_rule, PREDICATE_FORMULA
from n2s_lab.types import Extraction

print(PREDICATE_FORMULA)
raw = json.loads(EXTRACT_PATH.read_text())["extracts"]["T1"]["extract"]
ex = Extraction.from_dict(raw)
g = ground(ex)
r = evaluate_rule(g)
print("admin", ex.administration_status)
print("verdict", r.verdict.value, r.fired_rule)
print(g.to_dict())
PY
```

Expect T1 SATISFIED, A=B=C=D true, X false.

Send-back: paste the verdict line.

C2 — Nested ladder (four teaching notes)

Intuition. _apply(ex, "G9") runs G3...G9. Deeper includes earlier.

```
cd ~/staging/cxr-evidence-grounding-lab
../cxrlabs/faiss_gpu1/bin/python curriculum-mapping/snippets/replay_ladder.py I2 C4 T1 T2
```

Expect:

- I2: miss until G9 (planned → given)
- C4: CONTRADICTION from G4; selector G8
- T1: match at none
- T2: unrepaired UNCERTAIN

Send-back: four lines, one per id.

C3 — Leftover freeze (read-only)

Intuition. Goal 10 did not add a cell. It tagged leftover shapes.

```
cd ~/staging/cxr-evidence-grounding-lab
../cxrlabs/faiss_gpul/bin/python - <<'PY'
import json
from pathlib import Path
f = json.loads(Path("artifacts/n2s-gl0-leftover-census.json").read_text()["finding"])
print("n_miss", f["n_miss"])
print("ids_miss", f["ids_miss"])
print("new_cell?", f["shapes_that_justify_new_translate_cell"])
print("freeze", f["v1_freeze_recommended"])
PY
```

Send-back: the six ids and shapes_that_justify_new_translate_cell=[].

C4 — Viewer selftest + live page

Intuition. :8263 is the same ladder as C2, rendered.

```
cd ~/staging/cxr-evidence-grounding-lab
../cxrlabs/faiss_gpul/bin/python map_viewer_server.py --selftest
./scripts/run_map_viewer.sh
# open http://127.0.0.1:8263/
```

Optional curriculum UI:

```
./scripts/run_mapping_curriculum.sh
# open http://127.0.0.1:8264/
```

Do not run (not this walkthrough): ./scripts/run_g2_analog_fill.sh (GPU 7B), frozen-d patches, Dual_full rescore.

Send-back: C4 done plus HTTP 200 on / or selftest ok.

Claim hygiene

Say: mapping on Dual extracts, nested Translate, measured collateral.

Never say: auto-correct every LLM failure; :8260 chose the repair class; G3-G9 are six independent editors.